# XIAMEN UNIVERSITY MALAYSIA

## MID-TERM EXAMINATION

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|---|---|---|---|---|
| Course Code: | COM104 | | | |
| Course Name: | Engineering Maths (II) | | | |
| Question Paper Setter: | Lohgheswary A/P Nagarethinam | | | |
| Academic Session: | 2022/04 | Question Paper: | A ☑ | B ☐ |
| Total No. of Pages: | 2 | Time Allocated: | 2 hours | |
| Additional Materials: | - | | | |
| Apparatus Allowed: | - | | | |

**INSTRUCTIONS TO CANDIDATES**

1. Answer all questions.
2. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

**DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.**

(Student ID: _____ Full Name: _____ )

**Question 1 (25 marks, CLO2, PLO1)**

The network in Figure 1 shows a proposed plan for the traffic flow around a new park that will house the Liberty Bell in Philadelphia, Pennsylvania. The plan calls for a computerized traffic light at the north exit on Fifth Street, and the diagram indicates the average number of vehicles per hour that are expected to flow in and out of the streets that border the complex. All streets are one-way.

(a) How many vehicles per hour should the traffic light let through to ensure the average number of vehicles flowing out? [5 marks, C3]

(b) Assuming that the traffic light has been set to balance the total flow in and out of the complex, what can you say about the average number of vehicles per hour that will flow along the streets that border the complex? [20 marks, C3]

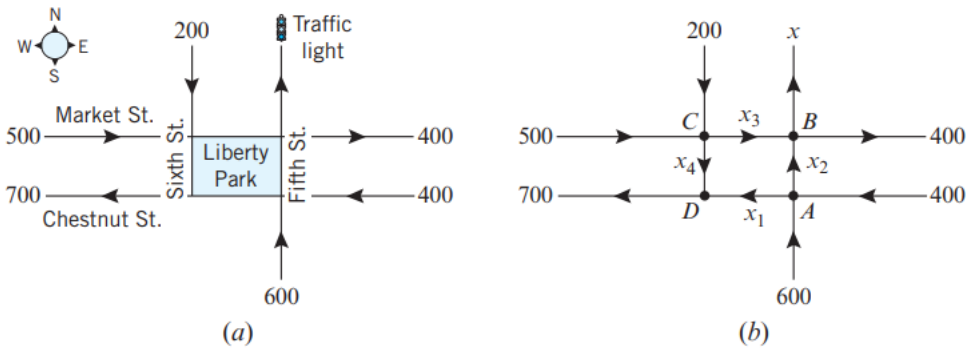



Figure 1

**Question 2 (25 marks, CLO2, PLO1)**

Use Cramer's rule to solve

$$x_1 + 2x_3 = 6$$
$$-3x_1 + 4x_2 + 6x_3 = 30$$
$$-x_1 - 2x_2 + 3x_3 = 8$$

**Question 3 (25 marks, CLO1, PLO1)**

The upward velocity of a rocket is given at three different times on the following table.

| Time, $t(s)$ | Velocity, $v(m/s)$ |
|---|---|
| 5 | 106.8 |
| 8 | 177.2 |
| 12 | 279.2 |

The velocity data is approximated by a quadratic polynomial as

$$v(t) = at^2 + bt + c, \quad 5 \leq t \leq 12.$$

(a) Set up the equations in matrix form. [9 marks, C3]

(b) Find the coefficients $a, b, c$ of the velocity profile by using the method of adjugate matrix. [16 marks, C3]

**Question 4 (25 marks, CLO1, PLO1)**

(a) Show that the vectors $\vec{v}_1 = (1,2,1)$, $\vec{v}_2 = (2,9,0)$, and $\vec{v}_3 = (3,3,4)$ form a basis in $\mathbb{R}^3$. [14 marks, C2]

(b) Find the coordinate vector of $\vec{v} = (5,-1,9)$ relative to the basis $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$. [9 marks, C2]

(c) Find the vector $\vec{v}$ in $\mathbb{R}^3$ whose coordinate vector relative to $S$ is $\vec{v}(s) = (-1,3,2)$. [2 marks, C2]

**- END OF PAPER –**